6-11

University of Rajshahi
(For Affiliated Colleges)
Department of Computer Science and Engineering
B.Sc. Engg. Part-1 Even Semester 2021
Course: STAT1211 (Statistics for Engineers)
(Session: 2020-2021)
Time: 2 Hours **Full Marks: 40**
[N.B. Answer FOUR questions taking at least TWO from each Section.]

Section A

1. (a) What are the measures of central tendency? Which one is best measure of central tendency? Why? 4

(b) Calculate mean, median and mode from the following distribution: 6

Marks	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80
No. of students	5	10	20	16	8	6

2. (a) Define probability, probability function and probability density function. 2

(b) State and prove Baye's theorem. 4

(c) In a bolt factory machines A, B and C manufacture respectively 40%, 10% and 50% of the total of their output 2%, 3% and 4% are defective bolts. A bolt is drawn randomly from the product and is found to be defective. What are the probabilities that it was manufactured by (i) Machine A? (ii) Machine B? (iii) Machine C? 4

3. (a) What do you mean by dispersion? What are the measures of dispersion? 3

(b) Define standard deviation and coefficient of variation. 3

(c) What is skewness? Discuss different types of skewness of a distribution with the help of diagrams. 4

Section B

4(a) Define outcomes, sample space and event of an experiment. 2

(b) For two observations, show that standard deviation is the half of the range. 4

(c) What is conditional probability? If $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.3$ then find $P(A|B)$ and $P(B|A)$. 4

5(a) What is scatter diagram? Interpret the values of r (correlation coefficient) with the help of scatter diagram. 3

(b) Mention some important properties of correlation coefficient. 2

(c) Given the following pairs of values of the variables X and Y 5

x	1	2	3	4	5
y	2	3	5	4	7

Compute the correlation coefficient (r) and comment.

6(a) What do you mean by statistical hypothesis? Differentiate the following pairs of concepts 6

(i) Parameter and Statistic;
(ii) Critical region and acceptance region;
(iii) Null hypothesis and alternative hypothesis.

(b) Stating the underlying assumptions, explain how you will test the quality of two population means. 4

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University of Rajshahi
(For Affiliated Colleges)
Department of Computer Science and Engineering
B.Sc. Engg. Part-1 Even Semester 2021
Course: STAT1211 (Statistics for Engineers)
(Session: 2019-2022)
Time: 2 Hours Full Marks: 35
[N.B. Answer FOUR questions taking at least TWO from each Section.]

Section A

1. (a) Which measures of central tendency is appropriate and why to find out the central value of the following observations- 22, 22.5, 21.5, 23, 22.9, 100 and 23.6? Show that mean deviation from arithmetic mean is zero. 2.75
(b) What do you mean by measures of dispersion? Define different measures of dispersion. 3
(c) Prove that standard deviation does not depend on origin but scale of measurement. 3

2. (a) Define random variable. How does it differ from variable? 1.75
(b) Let X be a random variable with pdf 3

$$f(x) = \begin{cases} kx, & 0<x<1 \text{ where } k \text{ is constant} \\ 0, & \text{otherwise} \end{cases}$$

Find the value of E(X) and V(X).
(c) Consider a random experiment in which two dices are tossed simultaneously. Compute sample space and find the probability that (i) sum of the points on the top of the dices is 6 (ii) greater than 2 (iii) less than 10 and (iv) equal to 10. 4

3. (a) What is kurtosis? What does it measure? 3
(b) Define skewness of a frequency distribution. 2.75
(c) Show graphically the approximate position of mean, median and mode when the distribution is (i) negatively skewed (ii) positively skewed and (iii) symmetrical. 3

Section B

4(a) What is correlation? Define correlation coefficient. Explain the values of correlation coefficient r when it takes values -1, 0 and +1. 4
(b) Consider the following data set on two variables x and y. 4.75

x	1	2	3	4	5
y	2	5	3	8	7

(i) Calculate the correlation coefficient between x and y.
(ii) Calculate the regression equation of y on x.

5(a) What do you mean by contingency table? What are the uses of such table? 2
(b) For the 2×2 contingency table prove that chi-square test of independence gives 3

$$X^2 = \frac{N(ad-bc)^2}{(a+c)(a+b)(c+d)(b+d)}, \qquad N = a+b+c+d.$$

(c) The following table shows price increase and decrease in market when credit squeeze is in operation or not. Use appropriate statistical test to find whether the credit squeeze has been effective in checking price rise using the critical value 3.29? 3.75

	Price decrease	Price increase
In operation	860	10
Not in operation	580	18

6(a) What do you mean by statistical hypothesis? Mention the various steps involved in testing a hypothesis. 3
(b) Define Type I and Type II errors. Stating the assumptions, explain how you can carry out test of significance 5.75
(i) $H_0: \mu = \mu_0$ $H_1: \mu = \mu_0$
(ii) $H_0: \mu_1 = \mu_2$ $H_1: \mu_1 \neq \mu_2$

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University of Rajshahi
Dept. of Computer Science and Engineering
B.Sc. Engg. Part-I, Even Semester, Examination 2020
Course Code: STAT 1211 Course Title: Statistics for Engineers

Time: 2 Hours **Full Marks: 35**

[N.B. Answer FOUR questions taking at least TWO from each Section.]

Section- A

1(a) Which of the given figure does present a descriptive statistics or an inferential statistics? Explain your answer. 3

Fig. 1(a)

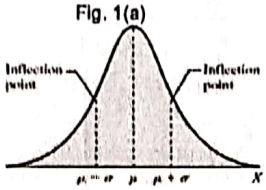
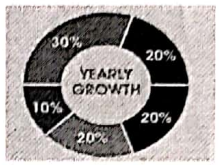


Fig. 1(b)



(b) The Chairman of the Dept. of Computer Science and Engineering wants to compare the academic performance between the students of session 2019 and 2020. He has the CGPA of all students and he wishes to get this comparison done by you since you have studied a course of Stat-1211. Explain the statistical approaches you may follow for this task. Mention pros and cons of those approaches if any. 2.75

(c) The probability distribution is shown by the given figures. What will be approximate position of mode, median and mean? Explain your answer. 3

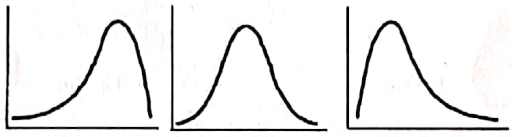


Fig.1(c) Fig. 1(d) Fig. 1(e)

2(a) What is central tendency? Write down the measure of central tendency. 2.75

(b) For non-zero positive observations show that $AM \geq GM \geq HM$ (Notations are usual). 3

(c) How can we measure the symmetry and asymmetry of a data distribution? Explain. 3

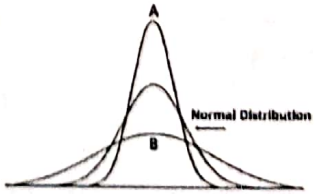
3(a) A box contains two **Red** balls. Another box of identical appearance contains one **Red** and one **White** ball. If a box is selected by chance and one ball is drawn from it, what is the probability that the **first box** was the selected one, if the drawn ball turns out to be **Red**? 4

(b) A bag contains **two white** and **four black** balls. Two balls are drawn. In $f(x, y)$, let x and y represent the results of the two drawings; 0 corresponding to a black ball, 1 corresponding to a white ball. Find, $f(0,1)$, $f(0,0)$, $f(1,0)$, $f(1,1)$, then from these four values, find $f(0)$ and $f(1)$. 4.75

Section-B

4(a) X denotes (i) the chance of **exact amount** rain tomorrow, (ii) **sum of the points** obtained in rolling two dices, which one is discrete and which one is continuous random variable? And why? 2

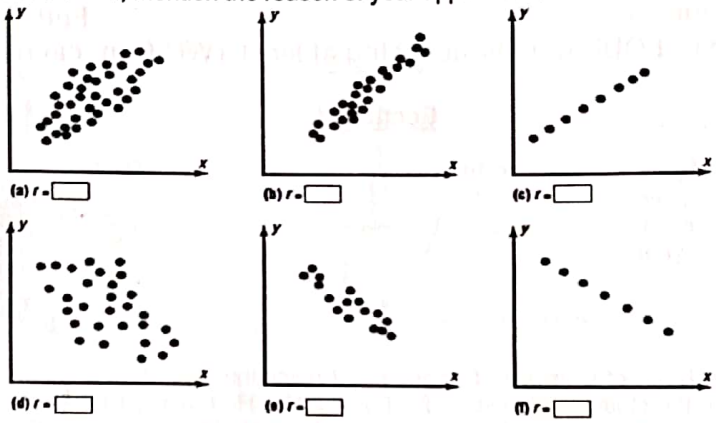
(b) For the distribution given in the figure, between A and B, which one is **platykurtic** and which one is **leptokurtic**? Explain your answer. 2



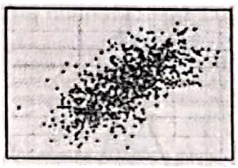
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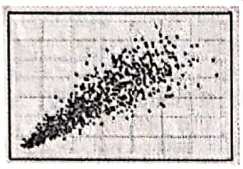
Marks: 35

(c) Interprete the distribution given below. Assign an approximate value of **Correlation Coefficient** for each of following distribution, mention the reason of your approximate values. 3



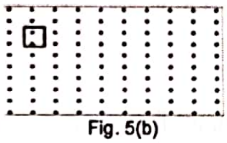
(d) Which of the given figures, does not comply Pearson Correlation and why? 1.75

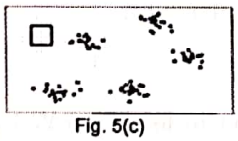

Fig 4(a)


Fig. 4(b)

5(a) Mention, which of the given distribution complies with the assumption of Poisson distributions or not. Explain your answer. 3


Fig. 5(a)


Fig. 5(b)


Fig. 5(c)

(b) A survey from Teenage Research Unlimited (Northbrook, Illinois) found that 30% of teenage consumers receive their spending money from part-time jobs. If 5 teenagers are selected at random, find the probability that at least 3 of them will have part-time jobs. 3.75

(c) The both two curves in each figure present the Normal distribution. But what is the difference between two curves in each figure? 2

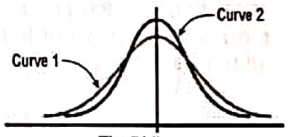


Fig.5(d)



Fig. 5(e)

6(a) Define null hypothesis and alternative hypothesis. 2.75

(b) Describe the test procedure for comparing two means. 3

(c) How can you test differences of variances of two populations? Is it possible to test variances using t-test? Which statistical test is applicable for test of independence? 3

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Department of Computer Science and Engineering

B.Sc. in Engineering 1st Year Even Semester Examination-2019

Course: **STAT-1211 [Statistics for Engineers]**

Marks: 35 [Answer any **four** (04) questions taking **two** (02) from each section.] Time: 02 Hours

Section-A

1.a) What is variable? Define with example different types of variable. 3

b) What do you mean by dispersion? Discuss different measures of dispersion 2¾

c) Calculate the standard deviation from the following data 3

40, 60, 65, 65, 65, 68, 68, 70, 70, 70, 70, 70, 70, 74, 75, 75, 90, 95.

$\bar{x} = \frac{\sum x}{n} = \frac{1260}{18} = 70$

$SD = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = 11.308$

2.a) Define outcome, sample space and event of an experiment. 3

b) State the law of total probability. 2

c) What is conditional probability? If $P(A) = 0.4, P(B) = 0.5$ *and* $P(A \cap B) = 0.3$, then find $P(A|B)$ *and* $P(B|A)$. 3¾

3.a) What do you mean by random variable? 2

b) Write down the probability density function of a normal distribution. What are the properties of standard normal distribution? 3¾

c) Given that X is a random variable whose mean $= 3$, find the variance of $4X + 2$. 3

Section-B

4.a) Define correlation coefficient, partial correlation coefficient and multiple correlation coefficient. 3

b) How do you interpret the range of correlation coefficient? 2¾

c) Find the value of the correlation coefficient from the following table: 3

Age (X)	43	21	25	42	57	59
Glucose Level (Y)	99	65	79	75	87	81

5.a) Explain the simple linear regression model. 2¾

b) Discuss the principles of least-square method. 3

c) Fit the regression equation from the following data with x as the independent variable. 3

X (aptitude test)	95	85	80	70	60
Y (Statistics grade)	85	95	70	65	70

$r = 0.529$

$\sum x = 247 \quad \sum x^2 = 11409 \quad \sum xy = 20485$

$\sum y = 486 \quad \sum y^2 = 40022$

6.a) Write down the goal and different steps of hypothesis testing. 2

b) Describe the test procedure for comparing two means. 4

c) The average daily intakes of dairy products for men (sample size=50, sample mean=756 and sample standard deviation=35) and women (sample size=50, sample mean=762 and sample standard deviation=30). Is there a difference in the average daily intakes of dairy products for men versus women? (At 5% level of significance, the critical region is $z > 1.96$ or $z < -1.96$) 2¾

For $\mu_1 = \mu_2$

$S_1^2 = 35^2 = 1225$

$S_2^2 = 30^2 = 900$

$z = \frac{756 - 762}{160\sqrt{\frac{1}{50} + \frac{1}{50}}}$

$S^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} = \frac{(50-1)1225 + (50-1)900}{50+50-1}$

$= \frac{60025 - 44100}{99}$

$= 160$

$z = \frac{\bar{x}_1 - \bar{x}_2}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} =$

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-1 Even Semester, Examination-2018
Course: STAT-1211 (Statistics for Engineers)

Time: 2 Hours | Full Marks: 35

Answer any **four** questions taking **two** from each Section

SECTION: A

1 (a) What is central tendency? Write down the measures of central tendency. 2.75

(b) For non-zero positive observations show that $AM \geq GM \geq HM$(Notations are usual). 3

(c) The daily profit of 100 shops in a market are distributed as follow: 3

Profits (in lac Tk)	50-60	60-70	70-80	80-90	90-100
No. of Shops	18	32	24	16	10

Calculate mean, median and mode.

2 (a) Define coefficient of variance (c.v). Why do you prefer coefficient of variation instead of standard deviation? 2.75

(b) For two observations, show that standard deviation is the half of the range. 3

(c) Show that variance is independent of origin but not of scale. 3

3 (a) Define probability, probability function and probability density function. 2.75

(b) State and prove Baye's theorem. 3

(c) Two unbiased dice are tossed simultaneously. What is the probability of getting a total of point 6 or even numbers from both the dice? 3

SECTION: B

4 (a) Define Binomial distribution. Write down few of its properties. 2.75

(b) Find mean, variance and coefficient of skewness(β_1) of Poisson distribution. 3

(c) If electricity power failures occur according to a Poisson distribution with an average of failures every twenty weeks, calculate the probability that there will not be more than one failure during a particular week. 3

5 (a) What is correlation? Write down the properties of correlation. 2.75

(b) Derive the correlation co-efficient and interpret it. 3

(c) If X and Y are independent then show that they are uncorrelated. 3

6 (a) Define null hypothesis and alternative hypothesis. 2.75

(b) Describe how will you test the null hypothesis $H_0: \mu = \mu_0$ vs. $H_1: \mu \neq \mu_0$. 3

(c) Distinguish between normal test and t-test. 3

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University of Rajshahi

Department of Computer Science and Engineering

B.Sc.(Engg.), Part-1, Even Semester Examination-2017

Course: STAT 1211 (Statistics for Engineers)

Time: 2 Hours **Full Marks: 35**

[Answer FOUR (04) questions taking TWO (02) from each Section.]

Section A

1. (a) What is meant by dispersion? What are the important measures of dispersion? 2.75
   (b) If $\bar{x}$ and s denote respectively the mean and standard deviation of a set of n observations, show that $\bar{x}\sqrt{(n-1)} \geq s$. 3
   (c) Calculate the mean and variance of the first n natural numbers. 3

2. (a) Define with example event, mutually exclusive event and sample space. 2.75
   (b) State and prove the additive law of probability for two events. 3.5
   (c) Three events A_1, A_2 and A_3 are mutually exclusive and their union is the sample space S. 2.5
   If $P(A_1) = \frac{3}{2}P(A_2), P(A_2) = 2P(A_3)$, then find $P(A_1)$, $P(A_2)$ and $P(A_3)$.

3. (a) Define binomial distribution. 2.75
   (b) If X is distributed as binomial, show that mean ≥ variance. 3
   (c) Let three unbiased coins are tossed at a time. What is the probability that (i) no head, (ii) at least two heads will appear? 3

Section B

4. (a) How do you distinguish between correlation and regression? 2.75
   (b) Show that correlation coefficient is independent of change of origin and scale of the variables. 3
   (c) Calculate the rank correlation coefficient from the following data on hourly sales (x) and expenses (y) of 10 stores 3

x	50	50	55	60	65	65	65	60	60	60
y	11	13	14	16	16	15	15	14	13	13

5. (a) What is a scatter diagram? What are the uses of scatter diagram? 2.75
   (b) Discuss the principles of least square method. 3
   (c) Estimate the parameters of the simple linear regression model using ordinary least square method.

6. (a) Define (i) parameter and (ii) level of significance. 2.75
   (b) Prices of shares in (Tk.) of a company on the different days in a month were found to be 66, 65, 69, 70, 69, 68, 71, 63, 64, and 68. Test whether the mean price of shares in the month is 65 or not (critical value at 5% level of significance is 2.26). 3
   (c) The mean yields (in gm) of two sets of plots and their standard deviation are given below. Test the hypothesis that whether the difference in the mean yields of the two set of plots is significant or not. 3

	Set of 40 plots	Set of 60 plots
Mean yield/Plot	1254	1243
Standard deviation	34	28

Department of Computer Science and Engineering
B.Sc. Engg. Part-I Even Semester, Examination-2016
Course: STAT-1211 (Statistics for Engineers)

Time: 2 Hours | Full Marks: 35

Answer any four questions taking two from each group

PART: A

1 (a) Define location and dispersion. Among the measures of location, which one is best and why? 2.25
(b) Let AM=arithmetic mean, GM=geometric mean and HM=harmonic mean. Then show that $AM \geq GM \geq HM$. 3.5
(c) Given below is the marks obtained by students in a certain subject. Compute AM, GM and HM of the marks. 3
10, 10, 35, 40, 42, 47, 49, 50, 51, 63, 63, 77, 78, 78, 80.

2 (a) Define event and probability of an event. 2
(b) State and prove Bayes theorem. 3
(c) In a bolt factory machines A, B and C manufacture respectively 25%, 35% and 40% of the total. Of their output 5, 4, 2 percent are defective bolts. A bolt is drawn randomly from the product and is found to be defective. What are the probabilities that it was manufactured by machine A, B and C? 3.75

3 (a) Define random variable, joint density function and marginal density function. For two constants a and b, prove that E(aX+b)=aE(X)+b 5
(b) Define normal distribution. Find its mean and variance. 3.75

PART: B

4 (a) Define Spearman's rank correlation coefficient. Show that Spearman's rank correlation coefficient can be expressed as $\rho_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$. 5
(b) The marks achieved by the 1st year students of CSE department in 2015 of their following courses Statistics and Mathematics as 3.75

Statistics	28	32	26	31	34	20	16	21	33
Mathematics	25	31	30	22	27	26	29	33	32

Find the rank correlation coefficient between the courses.

5 (a) Define null hypothesis, alternative hypothesis, simple hypothesis and composite hypothesis. 2.75
(b) Describe different steps for testing a single mean. 3
(c) What do you mean by type-I and type-II error? Among them which one is more serious and why? 3

6 (a) Define contingency table. For 2X2 contingency table show that 3.75

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(a+c)(c+d)(b+d)}$$

(b) How would you test the independence of hair color and eye color of persons from the following table at 5% level of significance where $\chi^2_{0.05} = 3.84$. 2.5

Eye color	Hair color		Total
	Black	Not Black	
Black	40		100
Not Black	50	50	
Total	90		200

(c) The diameters of steel shafts produced by a certain manufacturing process should have a mean of 0.255 inches. The diameter is known to have a standard deviation of ρ=0.0001 inch. A random sample of 10 shafts has an average diameter of 0.2545 inches. Set up an appropriate hypothesis on the mean and test at 5% level of significance, where $Z_{0.025}$=1.96. 2.5

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B. Sc. (Engg.) Part-I, Even Semester, Examination 2015
Course: STAT1211 (Statistics for Engineers)
Full Marks: 35 **Time: 2 Hours**

[Answer any four questions taking two from each group]

Part A

1. a) What are the characteristics of a good measure of central tendency? 2

   b) What is median? How do you calculate it for the grouped data? When do you prefer it to the arithmetic mean? 3.75

   c) If $\bar{x}$ and S denote the mean and standard deviation of $x_1, x_2, \ldots, x_n$, then show that 3
   $$\bar{x}\sqrt{(n-1)} \geq S.$$

2. a) What do you mean by dispersion? Define Coefficient of variation and explain why this is sometimes preferred to the standard deviation as a measure of dispersion. 4.25

   b) The following is the distribution of drug addicted people by age coming to the hospital for treatment: 4.5

Class interval Of ages	18-22	22-26	26-30	30-34	34-38
No. of people	7	10	18	8	7

   Calculate mean, mode and standard deviation.

3. a) What do you mean by probability of an event? 2

   b) State and prove the multiplicative law of probability for two events. 2.75

   c) Write down the probability mass function of binomial distribution. Let a random variable X follows binomial distribution. Find the mean and variance of X. 4

Part B

4. a) What do you mean by Pearson correlation coefficient? 2

   b) Prove that the coefficient of correlation between two variables is independent of the origin and scale of measurement of the variables. 2.75

   c) Suppose an engineer makes the following measurements on temperature (T) and pressure (P) within a closed tank containing a fixed amount of gaseous product. 4

T (K)	298	301	302	310	312	323	337	341	349
P (atm)	1.2	1.3	1.5	1.7	1.8	2.0	2.1	2.3	2.8

   Calculate the correlation coefficient between T and P and interpret it.

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$y_i = \alpha + \beta x_i + e_i, \quad i = 1, 2, ..., n$

c) Suppose the weight (X) in kilograms and the distance (Y) in meters required to stop from a speed of 20 miles per hour, for a random sample of 10 different models of similar motorcycles are given below: 5

Weight (X)	314	317	320	326	331	339	346	354	361	369
Distance (Y)	13.9	14.0	13.9	14.1	14.0	14.3	14.1	14.5	14.5	14.4

i). Draw a scatter diagram for X and Y. Does the relationship look linear?
ii). Fit the regression line of Y on X and interpret the slope and intercept.
iii). Predict the expected stopping distance for a motorcycle that's weight is 360 kilograms.

6. a) What do you mean by statistical hypothesis? Discuss the different steps of hypothesis testing. 3

b) Differentiate the following pairs of concepts: 3
i). Null hypothesis and alternative hypothesis.
ii). Simple hypothesis and composite hypothesis.
iii). Type I error and Type II error.

c) A random sample of 15 men from a large community yields a mean height of 66.4 inches. The value of S (standard deviation) calculated from the sample is 3.1 inches. Do the data support the hypothesis that the mean height of men in the community is 65.0 inches? ($t_{0.05,14}$=2.145) 2.75

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University of Rajshahi

Department of Computer Science & Engineering

B. Sc. Engineering Part-I Even Semester-2013

Course: STAT-1211 (Statistics for Engineers)

Marks: 35, Time: 2 hours

Answer Four questions taking Two from each part

Part-A

1.a) What do you mean by central tendency? Why do we study dispersion? What are the absolute measures ofdispersion? 2.75

b) For two positive observations x and y show that AM>= GM >=HM. When does the equality hold? 3

c) Calculate mean and variance using the given data. 3

Mid Value (x)	5	10	15	20	25	30	35	40	45	50
Frequency (f)	1	4	15	31	49	22	25	12	5	3

2.a) Define probability of an event with example. 2

b) State and prove the additive law for three not mutually exclusive events. 3.75

c) A card is drawn from a pack of 52 cards. What is the probability that the card is either spade or king? 3

3.a) Define a random variable. A join density function of two random variables X and Y is given below 3

$f(x,y) = k(2x+y)$; $2<x<6, 0<y<5$

$= 0$; otherwise

Find the constant k.

b) Define mathematical expectation. For two constants a and b, prove that 2.75

(i) $E(aX + b) = aE(X) + b$ (ii) $v(aX+b) = a^2V(X)$

c) Define Binomial distribution. Find the mean and variance of Binomial distribution. 3

Part-B

4.a) What is a scatter diagram? 2

b) What is coefficient of correlation? Show that correlation coefficient is independent of changes of origin and scale of measurement. 3.75

c) Calculate correlation coefficient between the heights (in cm) and weights (in kg) of 10 students given below: 3

Height	150	155	157	160	162	165	168	172	175	180
Weight	55	58	60	61	65	62	74	75	80	89

5.a) How do you distinguish between correlation and regression? 2

b) Discuss the concept of rank correlation. Show that Spearman's rank correlation coefficient is given by: 3.75

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}.$$

c) Estimate the regression equation of Y on X for the given data: 3

Family members (X)	4	7	3	8	5	4	6
Expenditure on food in Taka (Y)	160	80	75	90	140	80	150

6.a) Define (i) Statistical Hypothesis, (ii) level of significance and (iii) critical region. 2.75

b) Let $x_1, x_2, \ldots, x_n$ be a random sample come from normal distribution with mean μ and variance σ^2, then describe the test procedure of the hypothesis 3

(i) $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$, where σ is known

(ii) $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$, where σ is unknown

c) A random sample of 15 men from a community yields a mean height of 67.6 inches. The value of standard deviation from the sample is 3.4 inches. Do the data support the hypothesis that the mean height of men in that community is 65.8 inches? ($t_{0.05,14} = 2.145$) 3

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc.Engg.(CSE) 1st Year Even Semester 2012
Course: STAT1211 (Statistics for Engineers)
Full Marks: 35 Time: 03 hours
[N. B. Answer any FOUR questions taking at least TWO each part.]

Part-A

1. (a) What do you mean by Statistics? What are the graphical methods of representing statistical data? Discuss any one of them. 3.75

(b) What is central tendency? What are its measures? Write down the characteristics of a good measure of central tendency. 3

(c) For the following frequency distribution calculate median with graphically: 2

Class interval	13-15	15-17	17-19	19-21	21-23
Frequency	3	7	12	6	3

2. (a) Define with examples: equally likely outcome and complementary event. 3
If $P(A)=0.35$, $P(B)=0.75$ and $P(A\cap B)=0.20$, find the values of $P(A\cup B)$ and $P(A\cap B^c)$.

(b) What do you mean by conditional probability? State and prove the multiplicative law of probability. 3.75

(c) The probability that doctor D will diagnose Cancer correctly is 65%. The probability that a patient will die by his correct diagnosis is 30% and the probability of death by wrong diagnosis is 75%. A patient of doctor D, who had Cancer, died. What is the probability that his Cancer was diagnosed correctly? 2

3. (a) Distinguish between discrete and continuous random variables. Define probability density function (pdf) and cumulative distribution function (cdf) of a random variable. 2.75

(b) The random variable X has the following probability function: 3

x	1	2	3	4
$p(x)$	0.1	0.3	0.4	0.2

Find $P(X\geq 2)$, $P(X<3)$, $E(X)$ and $V(X)$.

(c) What is Poisson distribution? Find the mean and variance of this distribution and hence show that the mean and variance of Poisson distribution are equal. 3

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Part-B

4. (a) Define positive correlation, negative correlation and zero correlation. 3.5

If $y = -\frac{ax+c}{b}$ then prove the correlation coefficient between x and y is -1 if signs of a and b are alike and +1 if they are different. 3

(b) Show that correlation coefficient lies between -1 and +1 2.25

(c) The height (X) and weight (Y) of a person in a locality are given below:

X (in inches)	60	55	59	62	70
Y (in lbs.)	82	90	95	106	135

Find the correlation coefficient between X and Y and comment.

5. (a) What is the meaning of a regression line? Why are there two regression lines in a bivariate distribution? 1.75

(b) Let the linear regression line of Y on X be $Y_i=\alpha+\beta Y_i+\varepsilon_i$; $i=1,2,.....,n$. Discuss the estimation method of α and β. Also, show that correlation coefficient is the geometric mean of the regression coefficients. 4

(c) Fit a straight line of price on supply to the following data by applying the method of least squares: 3

Supply, X:	85	89	90	93	96
Price (in Tk.), Y	127	115	98	100	95

Also, find the predicted price when supply is 80.

6. (a) What do you understand by hypothesis testing of significance? What types of errors may occur in hypothesis testing? Explain. 3

(b) Let $X_1,X_2,.....,X_n$ be a random sample from a normal population with mean μ and variance σ^2. Describe the test procedure to test the hypothesis H_0: $\sigma^2=\sigma_0^2$ against H_1: $\sigma^2\neq\sigma_0^2$. 3

(c) From a random sample of 15 values we calculate an estimate 5.2 for the variance of the population. Does the result support the hypothesis that the population variance is 8.5? 2.75

[At 5% level of significance the critical values of the test statistic are 5.63 (lower) and 26.12 (upper)]

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University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engg.(CSE) 1st Year 2nd Semester 2011

Course: STAT 1211 (Statistics for Engineers)

Time: 3 Hrs. **Full Marks: 35**

[N.B. Answer FOUR questions taking at least TWO from each part.]

Part A

1.a) For two positive observations, show that $AH=G^2$, where A=Arithmetic mean, H=Harmonic Mean, and G=Geometric mean. 1.75

b) "For two observations, standard deviation is the half of the range", – prove. 2

c) Calculate the mean, standard deviation and co-efficient of variation from the following data: 5

Class interval	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	2	5	7	13	21	16	8	3

2.a) Define with examples (i) sample space, (ii) event, and (iii) probability of an event. 3

b) If A and B are two events then prove that P(AUB)=P(A)+P(B)-P(AB). 2.75

c) Three events A, B and C are mutually exclusive and their union is the sample space Ω. If P(A)=(3/2)P(B), P(B)=2P(C), then find P(A), P(B) and P(C). 3

3.a) Define mathematical expectation of a random variable with example. 2

b) Given the following density function of the random variable X: 4

$f(x)=2(1-x)$; $0<x<1$

Find (i) E(x), (ii) E(3+5x), and (iii) V(x).

c) Find the mean and variance of binomial distribution. 2.75

Part B

4.a) What is scatter diagram? Discuss how it helps us in understanding the nature of relationship between two variables. 2.75

b) Define simple correlation coefficient, r. What does it measure? Prove that correlation coefficient is independent of both of change of origin and scale. 3

c) Given SS(X)=755, SS(Y)=654 and SP(X,Y)=655. Find r_{XY} and comment. 3

5.a) What is regression? 2

b) Consider a regression model of Y on X. derive the least squares estimates of the model parameters. 3

c) To study the tensile strength of a certain type of wire, the following pairs of observations were recorded, where X is the diameter in cm and Y is the mass supported in kg/cm. 3.75

X : 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4

Y : 14 26 50 56 42 98 82 88 134 124

Fit the regression model of Y on X for the above data. Find the predicted value of Y for X=2.8.

6.a) Define the terms: statistical hypothesis, critical region, Type-I error and Type-II error. 3

b) Suppose that X is a normal variate with mean μ and variance σ^2. Describe the test procedure to test the following hypothesis 2.75

$H_0 : \mu=\mu_0$

$H_1 : \mu\neq\mu_0$

c) The heights of 10 males in inches of a given locality are as follows: 3

70.2, 67.1, 62.5, 69.1, 70.2, 64.0, 64.6, 66.3, 61.7, 68.9

Is it reasonable to believe that average height is equal to 65 inches (The tabulated value of the test statistic at 5% level is 2.262)?

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